

TrES-2b

R-1073

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_150422 · created 2026-07-30T14:15:43+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457134.84368 +/- 0.00034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1183 +/- 0.0091
Transit depth (Rp/Rs)^2	1.4 +/- 0.22 [%]
Orbital Inclination (inc)	84.86 +/- 0.48
Airmass coefficient 1 (a1)	1.0302 +/- 0.0092
Airmass coefficient 2 (a2)	-0.0208 +/- 0.0027
Scatter in the residuals of the lightcurve fit is	0.89 %
Best Comparison Star	#3 - [53, 353]
Optimal Aperture	3.8
Optimal Annulus	5.75
Transit Duration (day)	0.0863 +/- 0.0056

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1183 $\pm$ 0.0091 vs 0.1254 (deviation 0.53 σ)
Duration fit vs published	0.0863 d vs 0.0746 d (deviation 1.43 σ)
Mid-time O-C	-0.9 min
Depth SNR	8.9
Residual scatter	0.89 %

Light curve

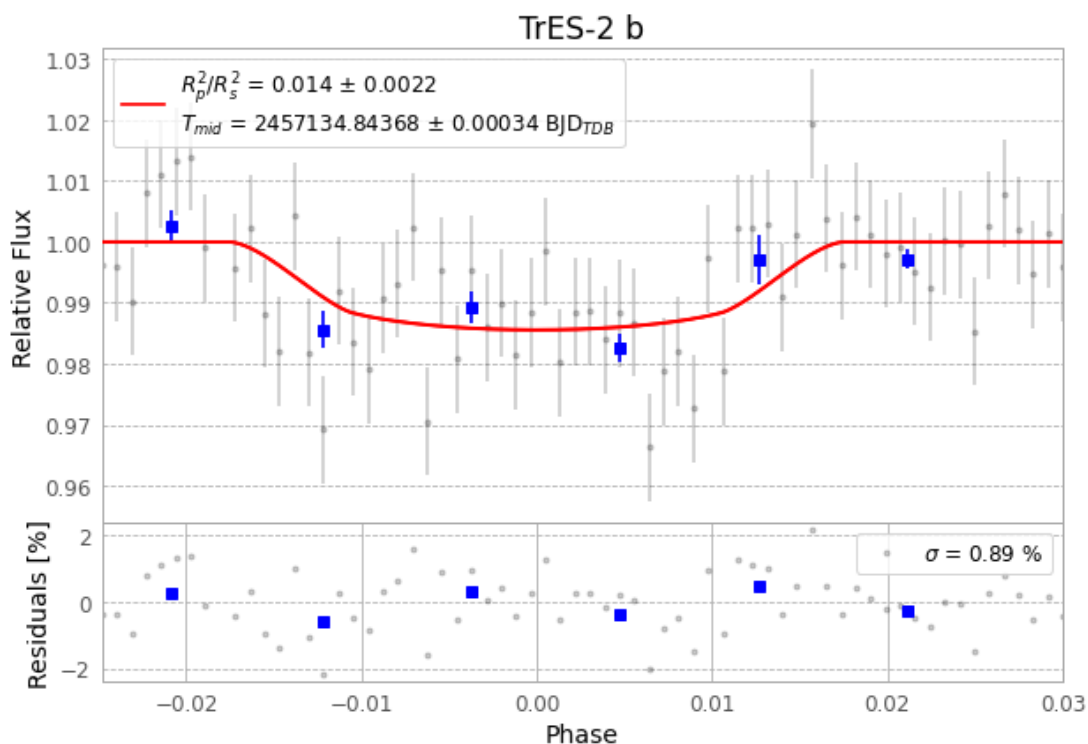


Figure 1 — FinalLightCurve_TrES-2 b_2015-04-21.png (SHA-256 5877a113be57c7f3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [191, 221], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b5652bf02d9d6e15...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

66 FITS frames (43 MB), TRES-2150422064512.FITS → TRES-2150422100012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest.

Artifacts: bench-TRES-2-150422.log (f9fcdffeea84...), FinalLightCurve_TrES-2 b_2015-04-21.png (5877a113be57...), FinalLightCurve_TrES-2 b_2015-04-21.csv (8782cf9bf0ef...)

Compute resources

Wall time 85.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 14:15Z.